3. Scope Item (3) – Treated Water Pipeline

3.1 Brief description and justification

All proposed pipeline routes were developed in accordance with Scottish Water regulations regarding land access and infrastructure installation requirements. Detailed route layouts are provided in Appendix C. Three alternative routing options were considered to transfer treated water between the water treatment works and the storage facility, each presenting different engineering, environmental, and operational characteristics. The three corresponding elevation charts with the addition of pump station are shown below respectively in Fig 1,2 and 3.

Route 1 primarily follows existing highway corridors, minimising disruption to private landowners and simplifying access for future maintenance activities. However, the route traverses more variable terrain and consequently requires six pumping stations to overcome elevation changes and maintain the required hydraulic performance.

Route 2 follows a combination of highway corridors and agricultural land. This alignment allows the pipeline to take advantage of more favourable topography, reducing the number of pumping stations required to four. The smoother terrain profile also enables greater utilisation of gravity-assisted flow, improving overall hydraulic efficiency and reducing operational energy demand.

Route 3 incorporates sections that cross major highways and pass through or near urban settlements, including towns and villages. While this route provides direct access to existing transport corridors, it introduces additional construction and stakeholder constraints associated with populated areas. Similar to Route 1, six pumping stations are required to satisfy the hydraulic requirements of the system.

For all route options, PE100 pipes with a diameter of 500 mm, a nominal pressure rating of PN10, and standard lengths of 12 m were selected. PE100 was chosen due to its favourable sustainability characteristics, corrosion resistance, long service life, and relatively low embodied carbon compared with traditional metallic pipe materials. Furthermore, the selected diameter provides efficient hydraulic performance by limiting frictional losses during water transmission. Preliminary hydraulic analysis indicates pipeline friction losses of approximately 0.37 m, 0.30 m, and 0.46 m for Routes 1, 2, and 3 respectively, excluding additional local losses associated with fittings, valves, bends, and other pipeline components.

The proposed pumping system utilises Lowara e-NSC pumps for the primary pumping stations. These pumps were selected due to their high hydraulic efficiency, maximum operating head of 160 m, and robust operational characteristics suitable for large-scale water transfer applications. Their PN16 pressure rating also provides an appropriate safety margin above the PN10 operating pressure of the selected pipeline. Clarke booster pumps were selected for similar reasons, offering reliable performance, operational flexibility, and compatibility with the overall system requirements.

3.2 Figures and tables

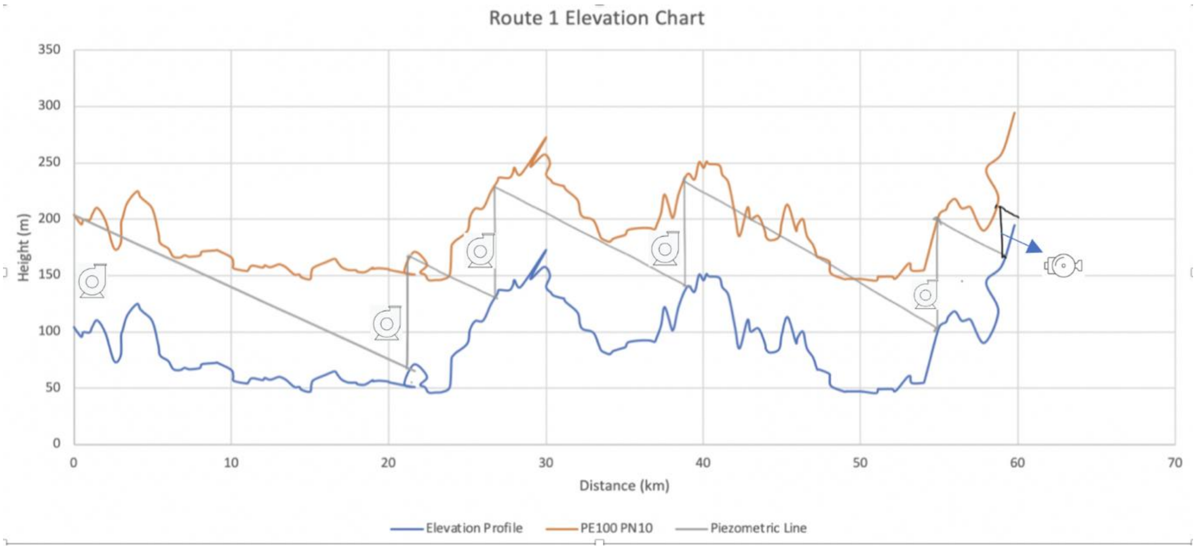

Figure 1: Elevation chart of routing option 1

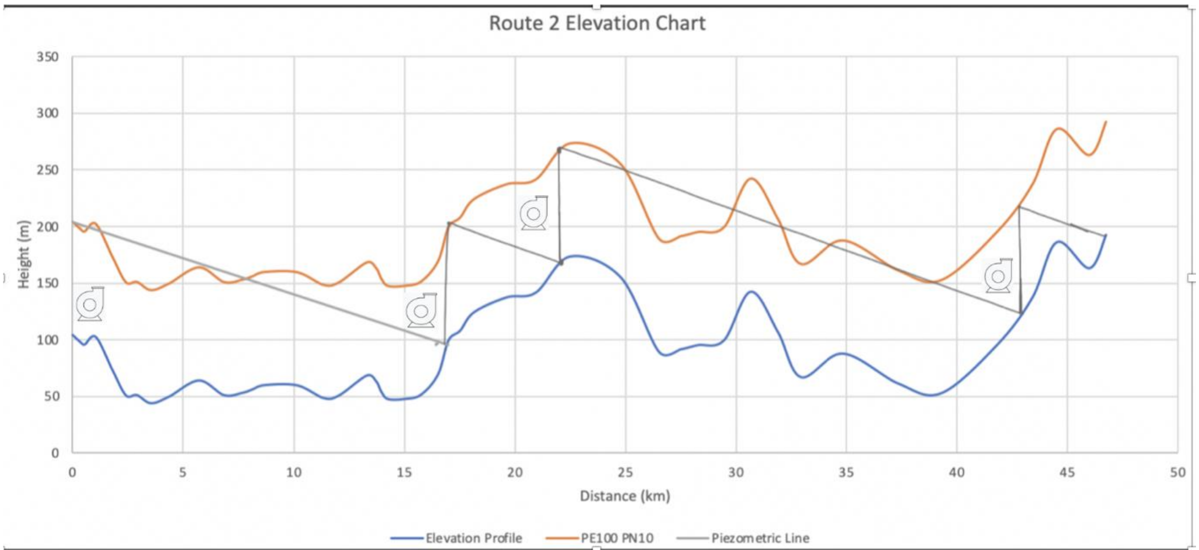

Figure 2: Elevation chart of routing option 2

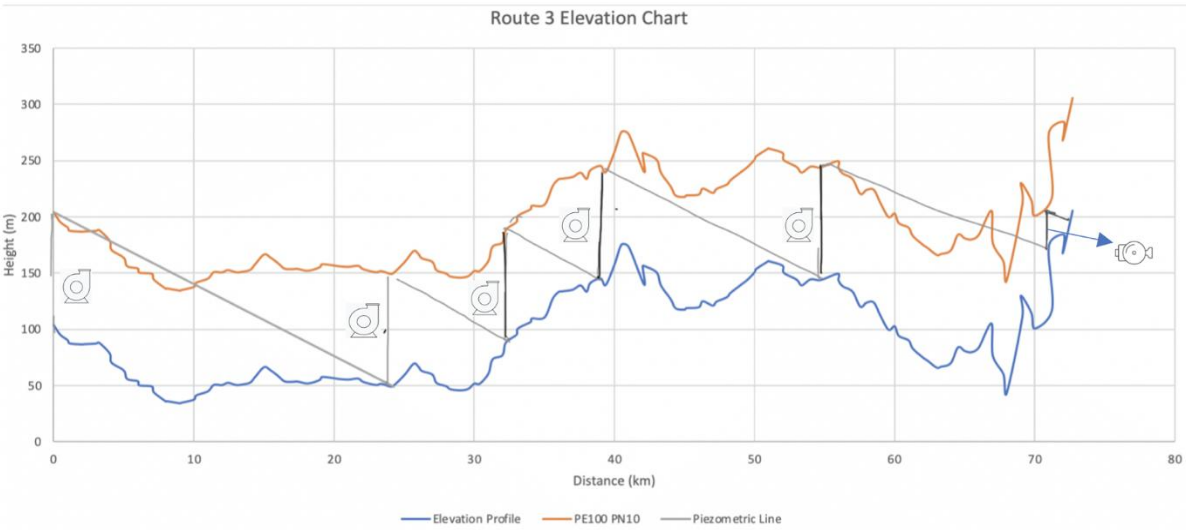

Figure 3: Elevation chart of routing option 3

	Route 1	Route 2	Route 3
Number of PE100 (500 mm, 12m length, PN10) pipes	Roughly 5000	Roughly 3917	Roughly 6100
Number of Lowara e- NSC pumps	5	4	5
Number of Clarke Booster Pump BPT1200SS 230v 7237006	2	0	1

Table 1: Comparison of the pump sizing of three different routing options

3.3 Recommendation

Based on the technical, environmental, and operational assessments, Route 2 is identified as the preferred option. This route demonstrates the lowest estimated carbon footprint associated with both pipeline and pumping infrastructure due to the reduced quantity of equipment required and shorter transportation requirements. In addition, Route 2 exhibits the lowest pipeline friction losses among the three alternatives and requires fewer pumping stations, resulting in lower operational energy consumption and reduced maintenance requirements.

The relatively smooth terrain profile along Route 2 maximises the use of gravity-assisted flow, improving overall hydraulic efficiency while providing greater operational flexibility. Collectively, these advantages contribute to lower whole-life carbon emissions, reduced operating costs, and enhanced long-term sustainability, making Route 2 the most suitable solution for the proposed treated water transmission system.

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Signature: